Adjusting method of DDR and DDR-T by servo guide

*Please refer to FANUC SERVO GUIDE manual (B-65404EN/01) for the servo guide method of popular use.

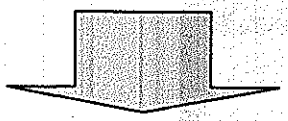
Contents

1. Flow of servo adjustment
2. Initial Gain-Tuning
3. Filter-Tuning
4. Gain-Tuning

1.Flow of servo adjustment(1/1)

The communication setting is done.

Setting and connection of NC, PC, and servo guide



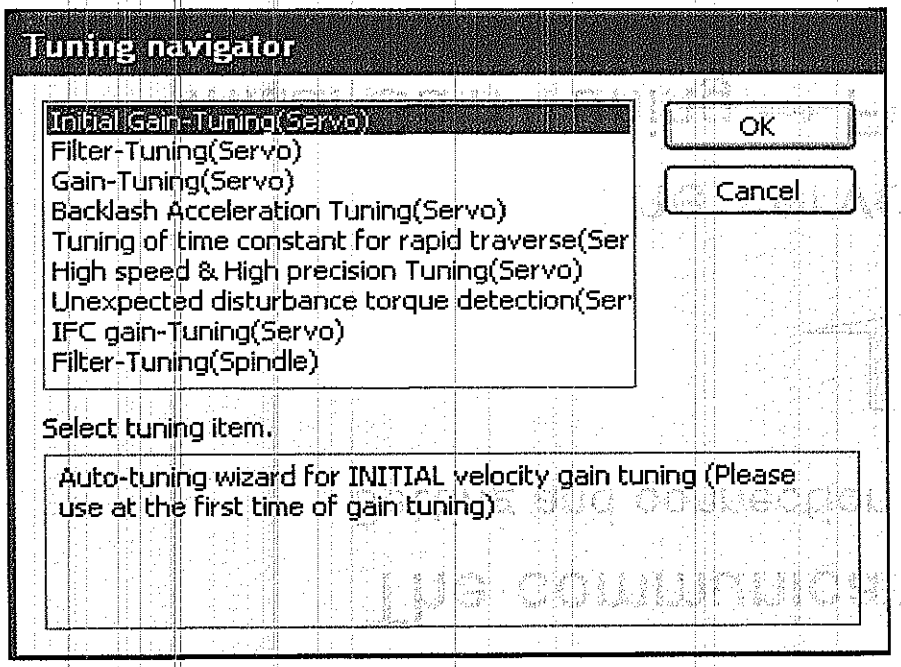
The servo is adjusted.

Initial Gain-Tuning → Filter-Tuning → Gain-Tuning

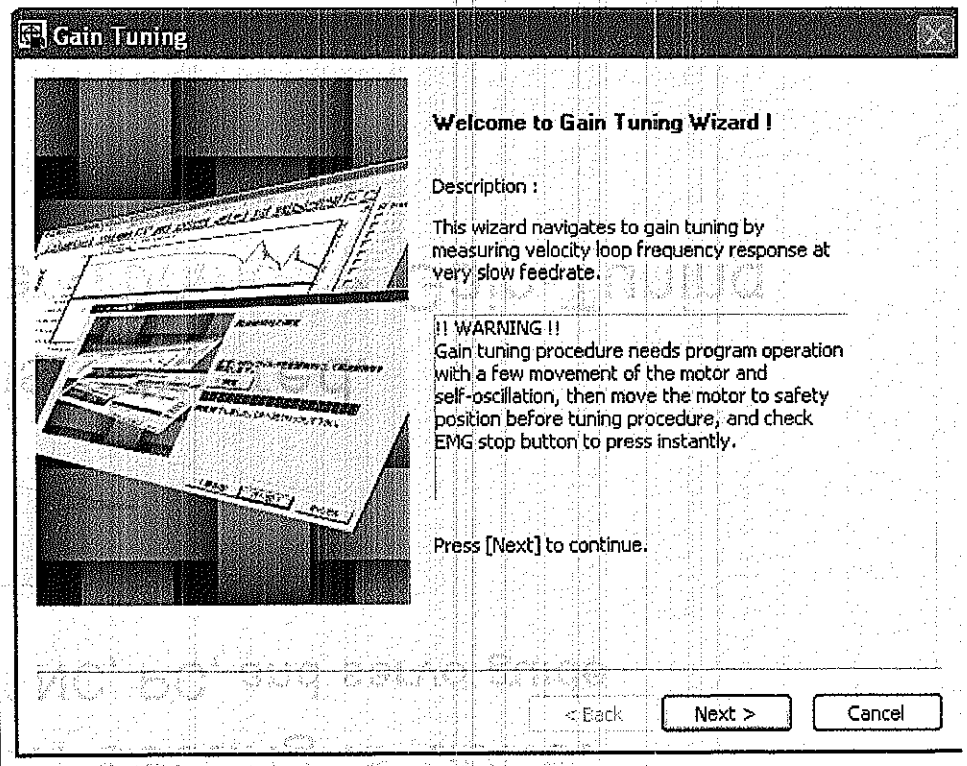
2.Initial Gain-Tuning(1/7)

Parameter | Graph | Program | Navigator... | Comm...

"Navigator..." of the servo guide is clicked.

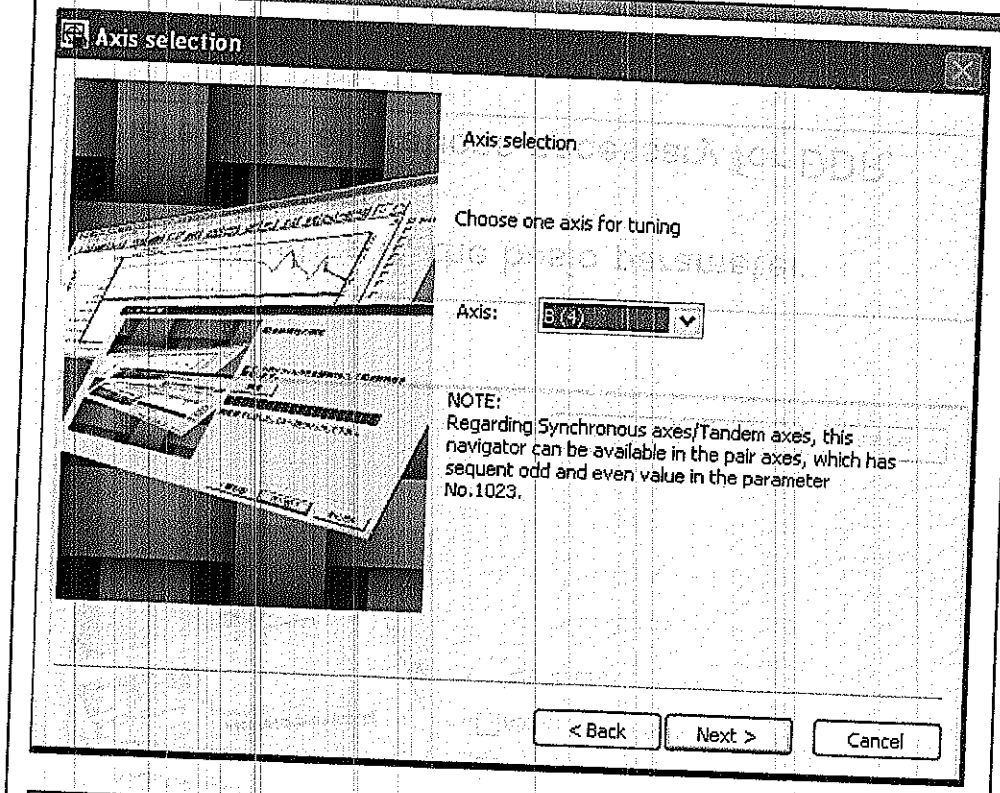


"Initial Gain-Tuning (servo)" is chosen, and "OK" is clicked.

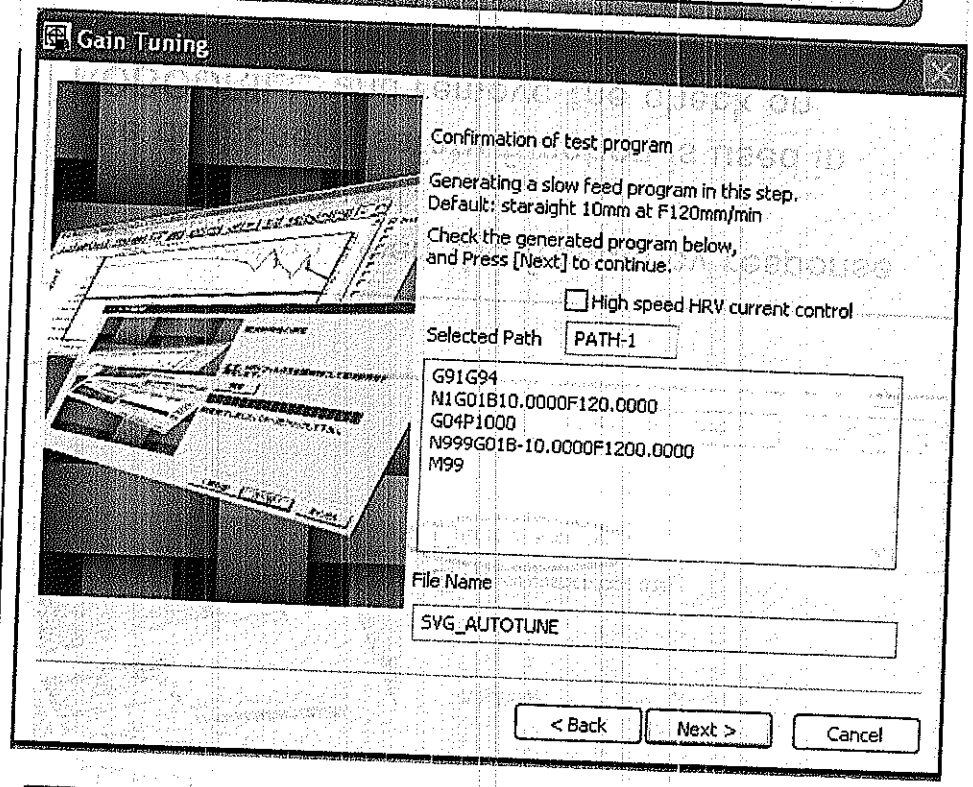


The explanation is displayed.
Move the motor to safety position before tuning procedure. Check the emergency stop button to press instantly.

2.Initial Gain-Tuning (2/7)

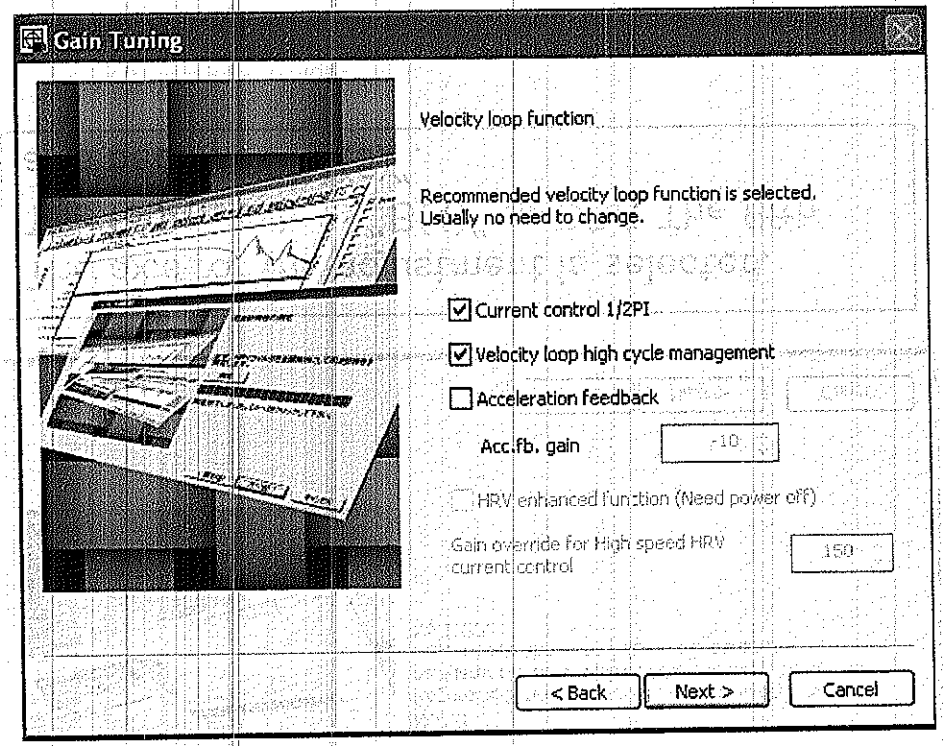


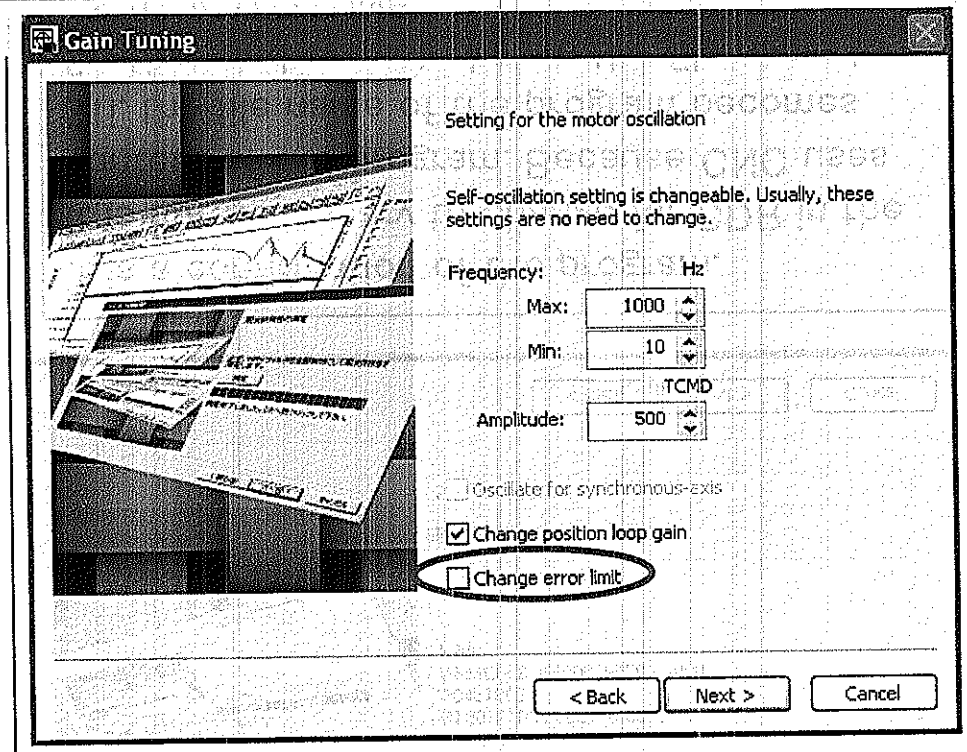
The axis for the adjustment is selected.
The fourth axis is "B(4)" at DDR The fifth axis becomes "A(5)".



It is a confirmation of the program.
It is not necessary to change for DDR in the content of the program. Because CNC uses 31i, the file name of the program becomes fixation, and cannot change the file name to "SVG_AUTOTUNE".

2.Initial Gain-Tuning (3/7)

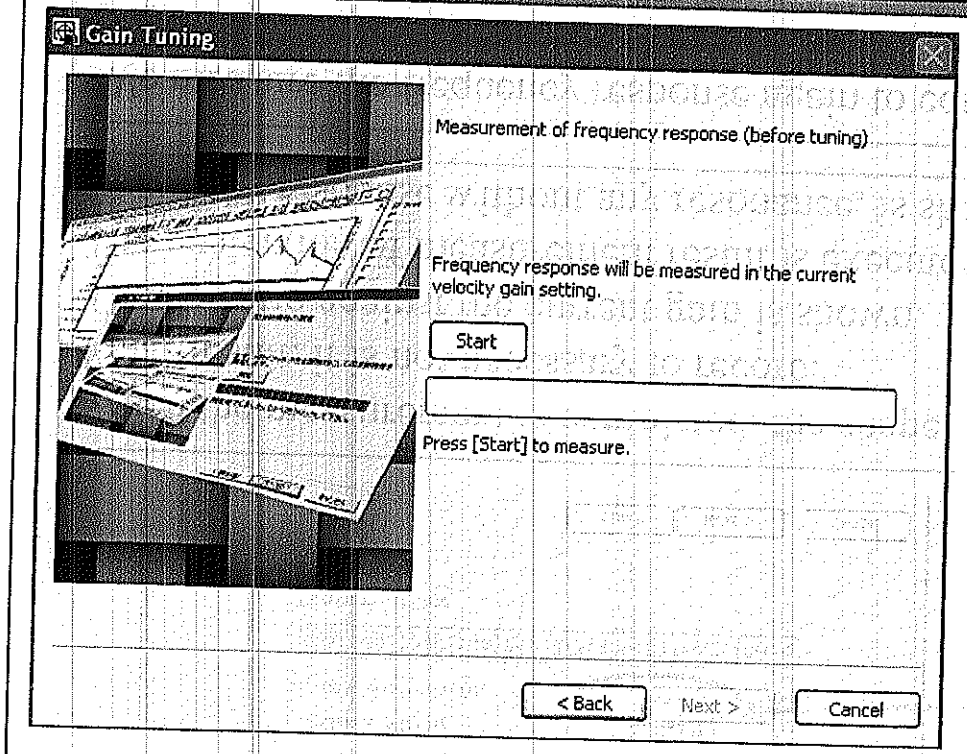




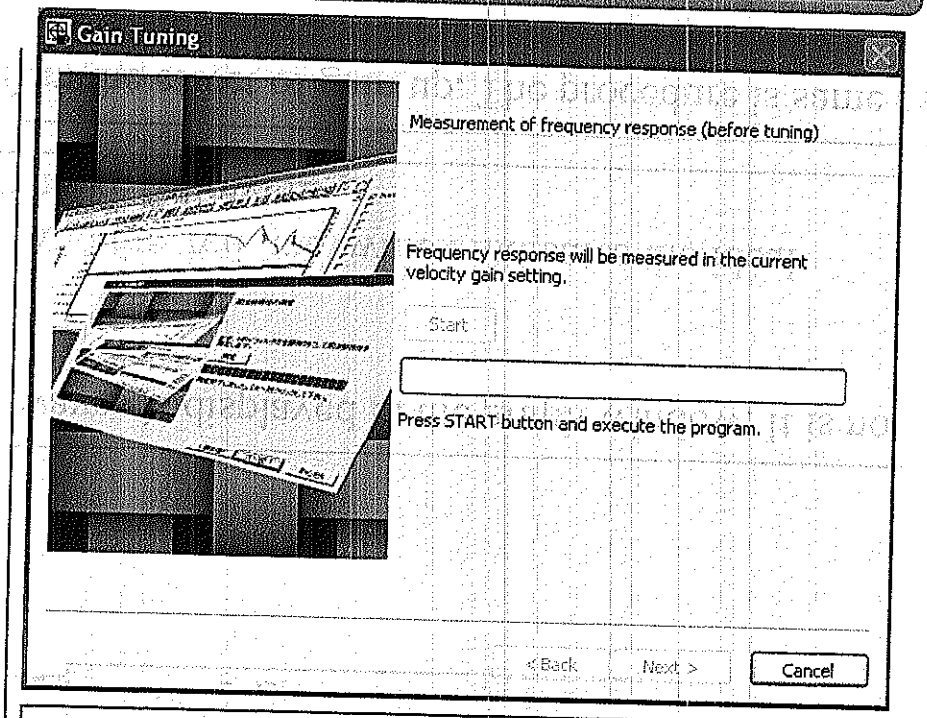
When adjusting it, the basic parameter condition is set.
There is no alterations necessary for DDR.

The condition when the frequency response is measured is set.
The dual check safety function is used in ROBODRILL, and remove the check on "Change error limit".
Other alterations necessaries do not exist.

2.Initial Gain-Tuning (4/7)

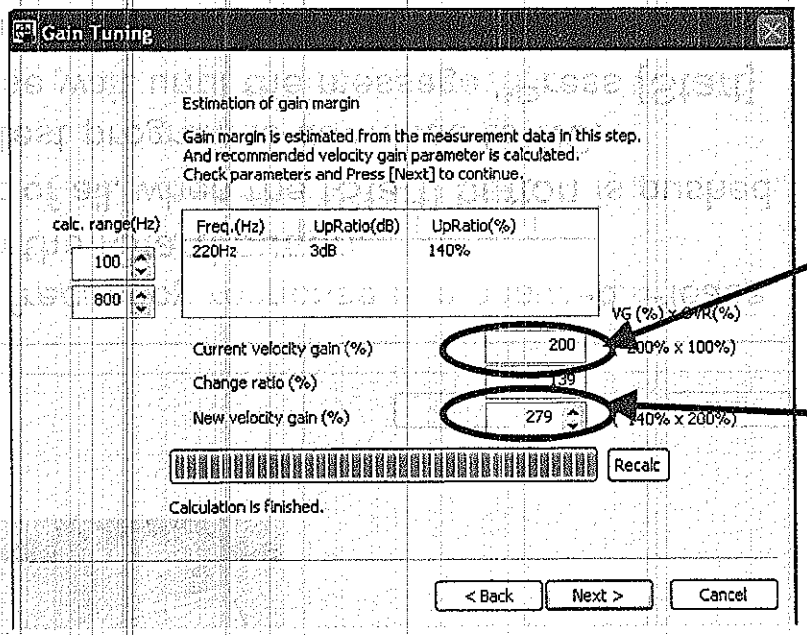


The frequency response is measured. Please push the [Start] button.
First of all, when the [Start] button is pushed, the test program is download to NC.
Please wait until the message "Press [Start] to measure " is displayed.



The machine works when cycle start button of the machine is pushed, and data acquisition starts. Please wait until the message " Measurement is completed" appears.
It might not go well by the relation such as triggers. Please do the measurement over again [Start] at that time.
*It is necessary to remove a single block.

2.Initial Gain-Tuning (5/7)



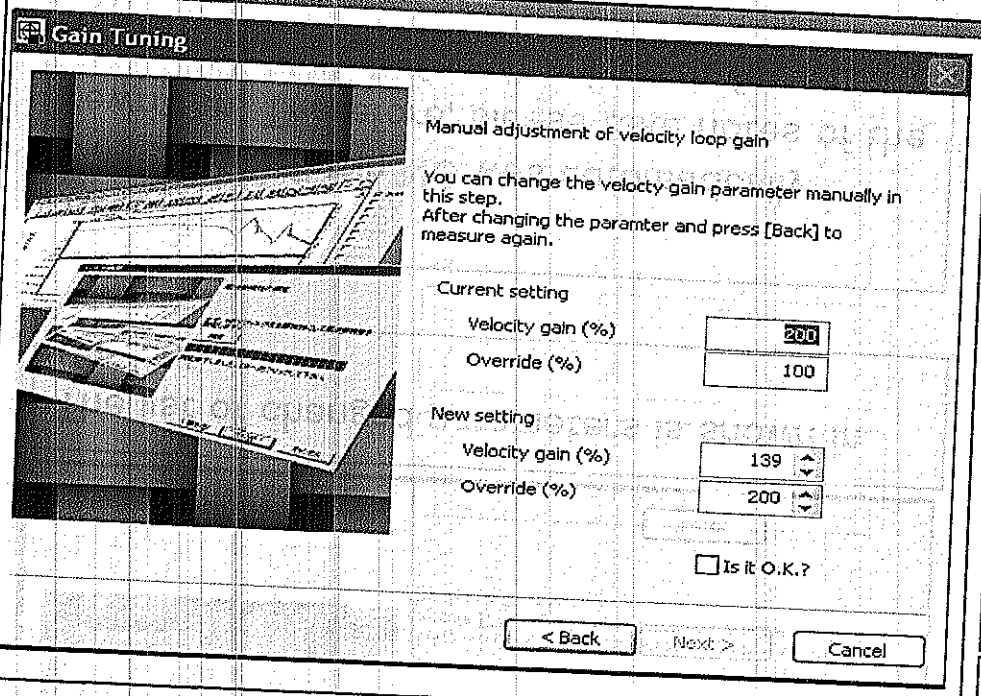
Velocity gain before adjusts it

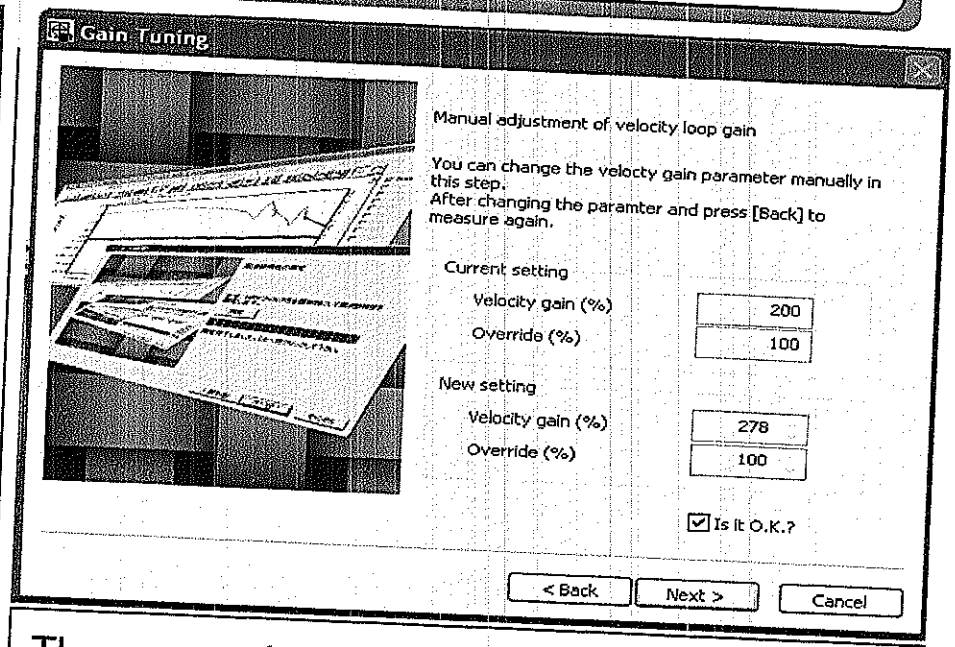
Velocity gain after adjusts it

When the measurement is completed, the shape of waves is displayed in the graph window. It is not Final-result, it is not necessary to record.
The up-ratio against the current gain is shown.
The evaluation of measurement result is executed. And the red line, which indicates the ideal frequency response without any resonance, is shown like following .

We measure the frequency response again to confirm the effect of gain up. The procedure is same as page 7 =2.Initial adjustment of gain(4/7)

2.Initial Gain-Tuning (6/7)





The final setting value of parameter is shown.
"Velocity gain" is used for rapid-traverse.
"Velocity gain × Override" is used for cutting.

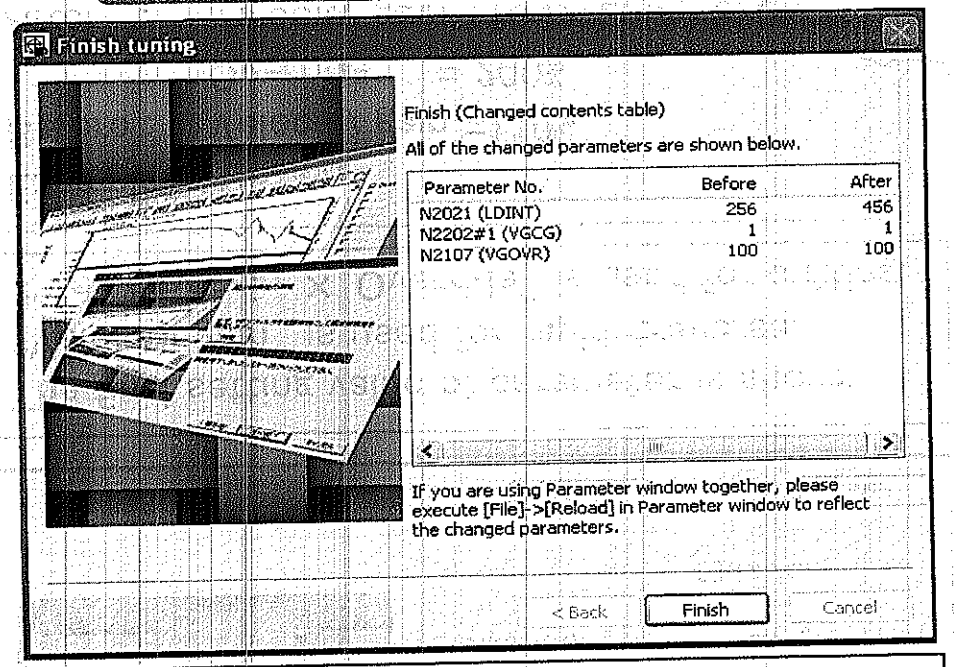
Example:

Before : Velocity gain =110%
Override = 200%

After : Velocity gain 110% × 200%=220%
Override 100%

The processing in the state of clamping is usually assumed in DDR. Therefore, it is thought the use only of rapid-yraverse. Therefore, the hand is corrected so that "Velocity gain × Override value" may become "Velocity gain". Moreover, to avoid oscillating, " Override " value is corrected to "100". "Is it O.K.?" is checked.
When you click "Next > ". The value of NC is rewritten.

2.Initial Gain-Tuning (7/7)



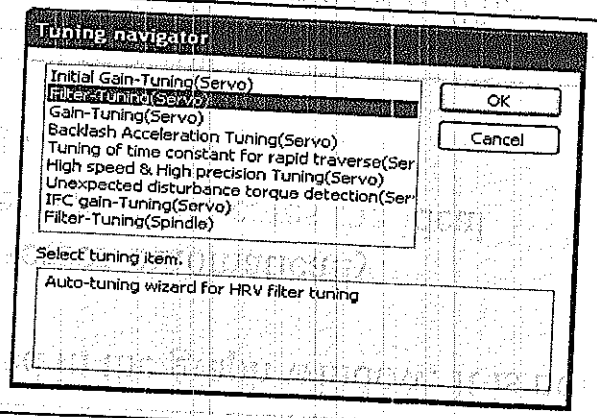
The list of changed parameters is shown in the last.

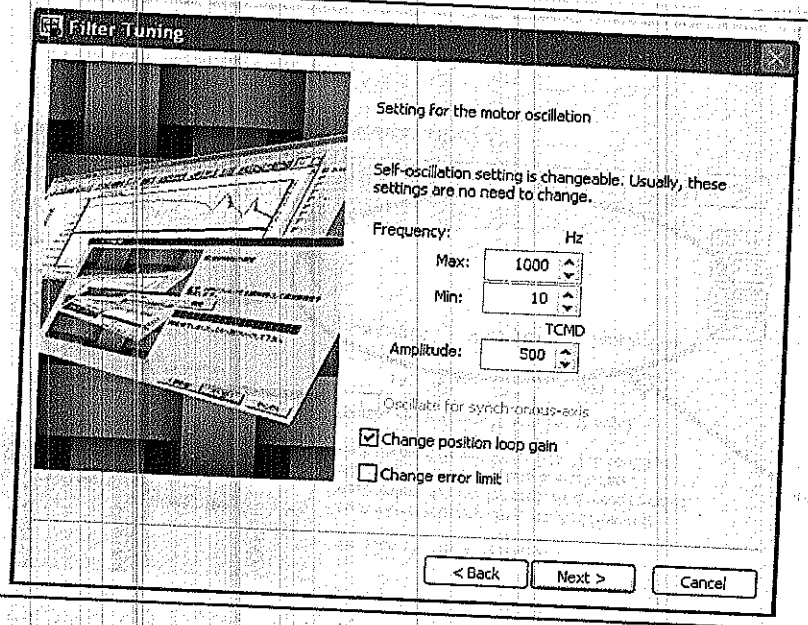
Attention:
The parameter in NC was automatically rewritten. Therefore, please take notes of the parameter data before it adjusts it for attention.

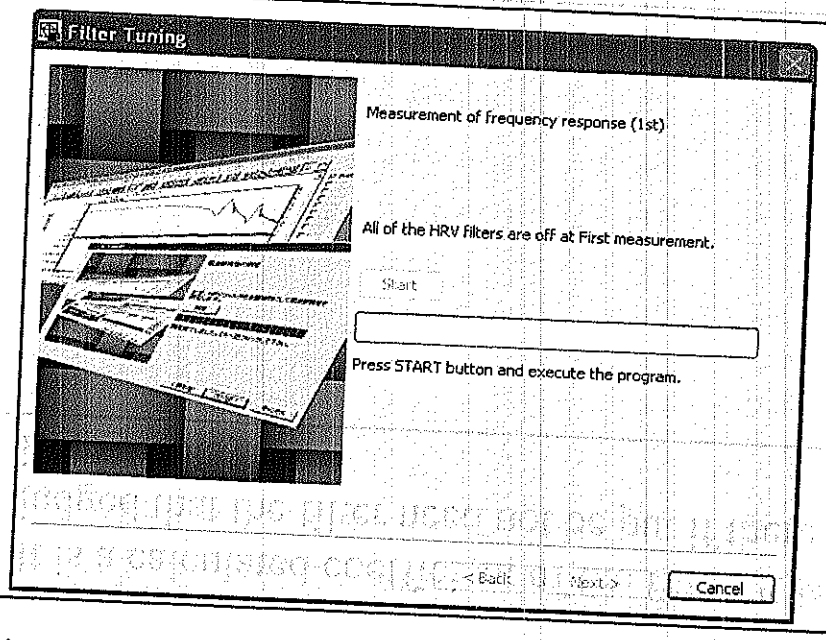
Reference:
At this point, it is a comparatively low gain moreover because it doesn't adjust the HRV filters (Resonance elimination filters).

3.Filter-Tuning(1/3)

- "Navigator…" of the servo guide is clicked. "Filter-Tuning (servo)" is chosen, and "OK" is clicked.
- Even the condition setting of the Measurement of frequency response (To P6) is the same procedure as "Initial Gain-Tuning (SERVO)".

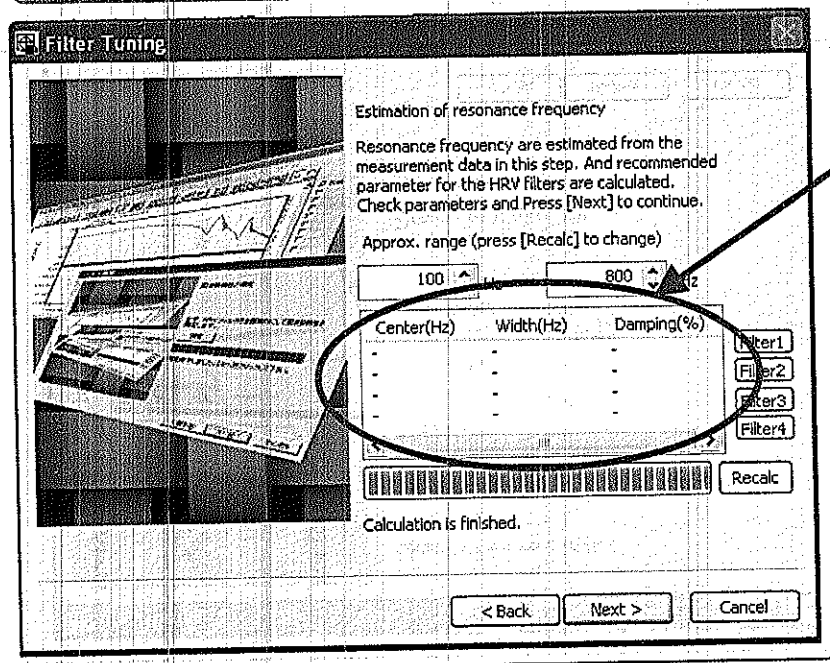






It is the same procedure as "Initial Gain-Tuning (SERVO)".
Though the display of the screen is different.
The frequency response is measured.

3.Filter-Tuning(2/3)

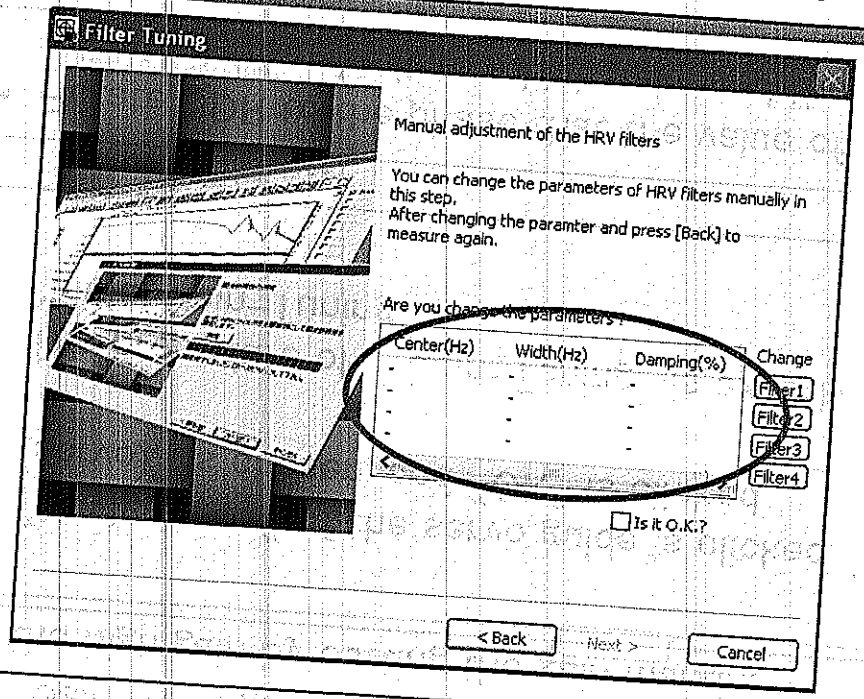


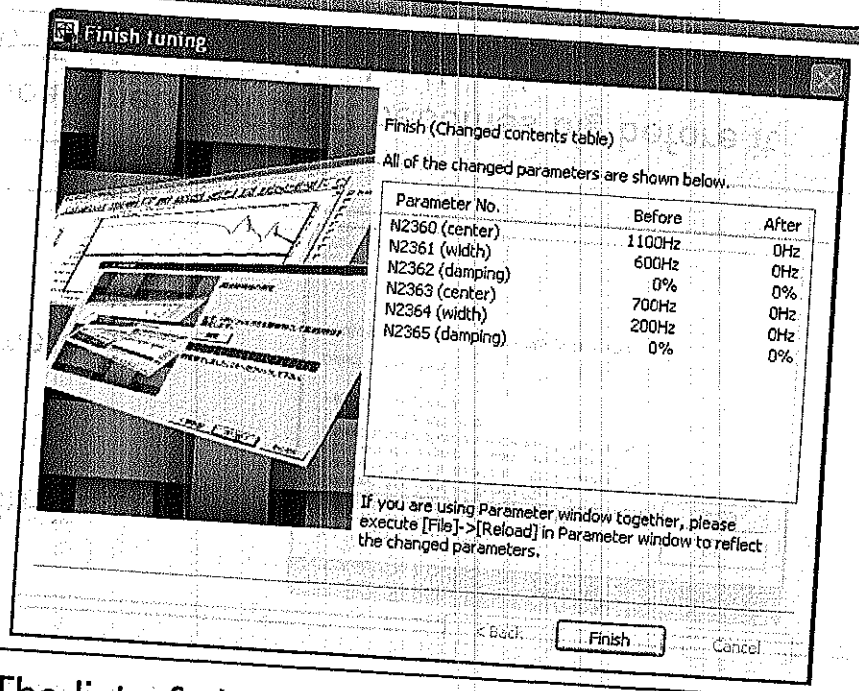
It is a calculated coefficient of the filter. It is judged that the filter need not be put if there is no display.

When the measurement is completed, the shape of waves is displayed in the graph window. It is not Final-result, it is not necessary to record.
The coefficient of the filters are calculated on the filter adjustment screen automatically.
The evaluation of measurement result is executed. And the red line, which indicates the ideal frequency response without any resonance, is shown like following .

The frequency response after the calculated coefficient of the filter is applied is measured. The content of work is the same as "Measurement".

3.Filter-Tuning(3/3)





After the frequency response result measured after the application of the filters is checked, the filters can be fine-tuned manually.
When the result satisfy you check "Is it O.K." and press <Next > button. The value of NC is rewritten.

The list of changed parameters is shown in the last.

Attention:
The parameter in NC was automatically rewritten. Therefore, please take notes of the parameter data before it adjusts it for attention.

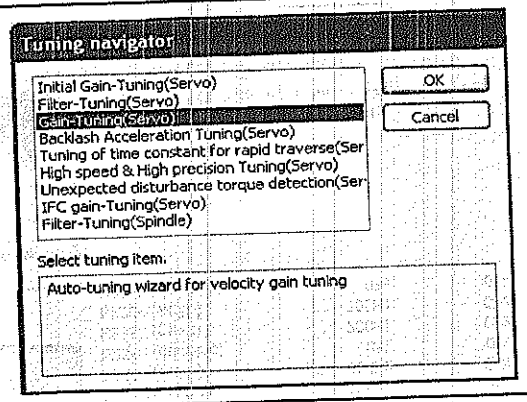
4.Gain-Tuning(1/4)

- There are two types of Gain tunings. One is "Initial Gain-Tuning" and the other is "Gain-Tuning ". In "Initial Gain-Tuning" , the velocity gain is determined with large margin against the limit of vibration level. When you use this tuning, the gain become a little larger than initial state, and the resonance of the machine becomes more clear.
  Next, we reject the resonance by Filter-Tuning.

- "Gain-Tuning. "is used for final determination without any resonance. This function realizes the high gain by cutting the gain margin.

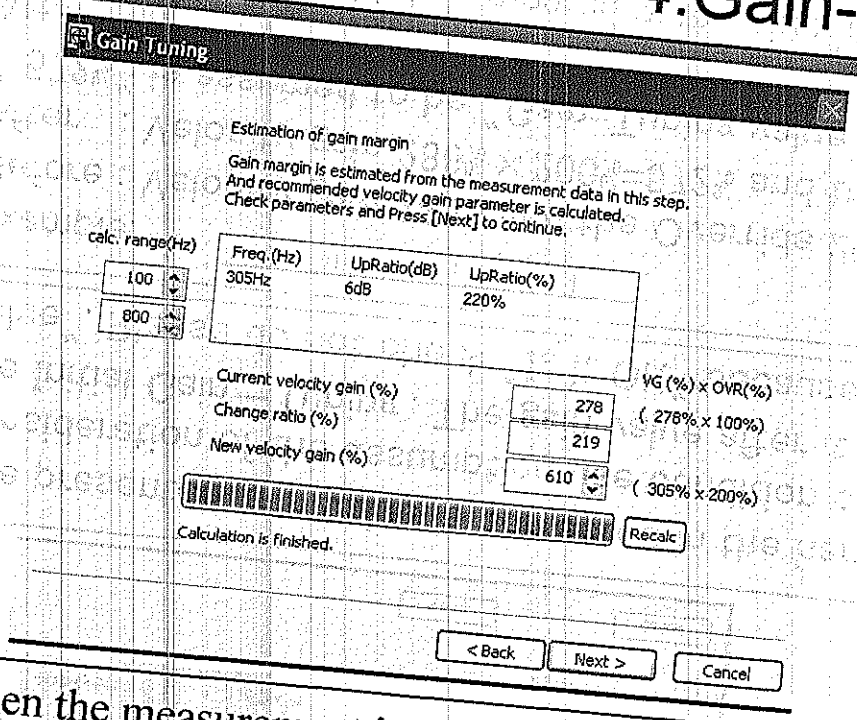
- "Navigator…" of the servo guide is clicked. "Gain-Tuning (servo)" is chosen, and "OK" is clicked

- The procedure of "Gain-Tuning" is exactly same as "Initial Gain-Tuning".



- Rotate DDR to the phase that the value of the load meter of DDR becomes big before it adjusts it to make it to a high gain, and the Gain-Tuning executes by the phase.

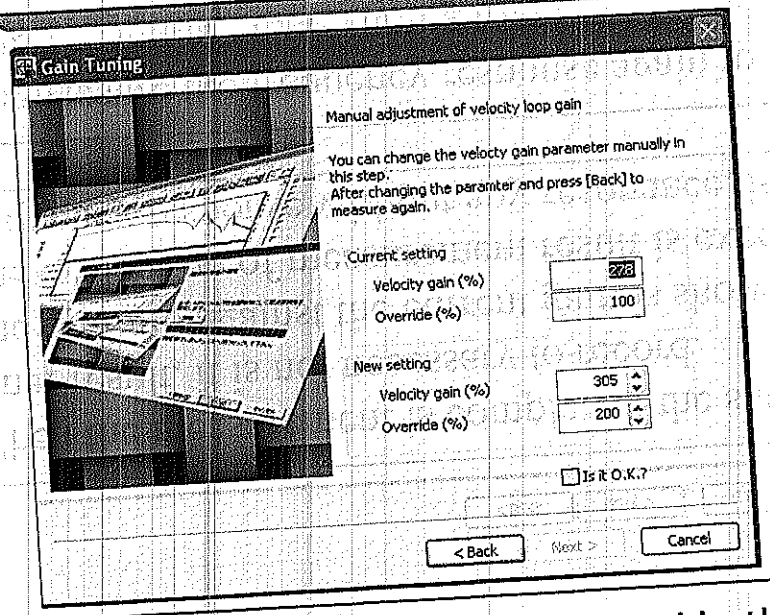
4.Gain-Tuning(2/4)



When the measurement is completed, the shape of waves is displayed in the graph window. It is not Final-result, it is not necessary to record.
The up-ratio against the current gain is shown.
The evaluation of measurement result is executed. And the red line, which indicates the ideal frequency response without any resonance, is shown like following .

We measure the frequency response again to confirm the effect of gain up. The procedure is same as page 7 = Initial Gain-Tuning (4/7)

Gain-Tuning(3/4)



The presented gain value is corrected in the hand as well as the Initial Gain－Tuning in consideration of the assumption use condition of DDR. The correction procedure is the same as the Initial Gain－Tuning . The gain value after it corrects it is assumed to be "Gain-Tuning value". Please do not check "Is it OK" because it is not the final gain value.

Example:
Before : Velocity gain 286% and the Override 200%
After : Velocity gain 286% × 200%=572% and the Override 100%.
" 572%" is assumed to be "Gain-Tuning value". (It is not the final gain value.)
Attention:
It is a considerably high compared with value in which the gain initial is adjusted a gain value. It doesn't use it as it is.

4.Gain-Tuning(4/4)

It compares the gain value of Gain-Tuning with the gain value of Initial Gain-Tuning.

80 % value of Gain-Tuning > value of Initial Gain-Tuning. .

80 % of the Gain-Tuning value is set as the gain value.

Example:
Gain self adjustment value =500%
80 % of the Gain-Tuning value =500 % × 0.8=400%
Value of Initial Gain-Tuning.=350%
80 % of the Gain-Tuning value is larger than value of Initial Gain-Tuning .We set 400% in velocity gain.

80 % value of Gain-Tuning < value of Initial Gain-Tuning.

The middle value of Gain-Tuning value and Initial Gain-Tuning value is set as the gain value.

Example:
Gain self adjustment value =400%
80 % of the Gain-Tuning value
=400 % × 0.8=320% Value of Initial Gain-Tuning.=350%
80 % of the Gain-Tuning value is smaller than value of Initial Gain-Tuning .We set the middle value of Gain-Tuning value and Initial Gain-Tuning value.

$$\text{Value of velocity gain} = \frac{\text{Value of initial gain-tuning} + \text{Value of gain-tuning}}{2}$$

$$= \frac{350\% + 400\%}{2} = 375\%$$

It is confirmed to run, and not to oscillate.